

Week 3 – Password Cracking Project (NetworkWalks Cybersecurity Internship, Batch B083)

Overview

This repo documents completion of the two Week 3 essential modules from the NetworkWalks Cybersecurity Internship: cracking password-protected PDF files using **John the Ripper (JTR) / Johnny GUI** and using **NetworkWalks' own online Hash Calculator + Password Cracker tools**.

Modules Completed

W3-PM1: Password Cracking with JTR

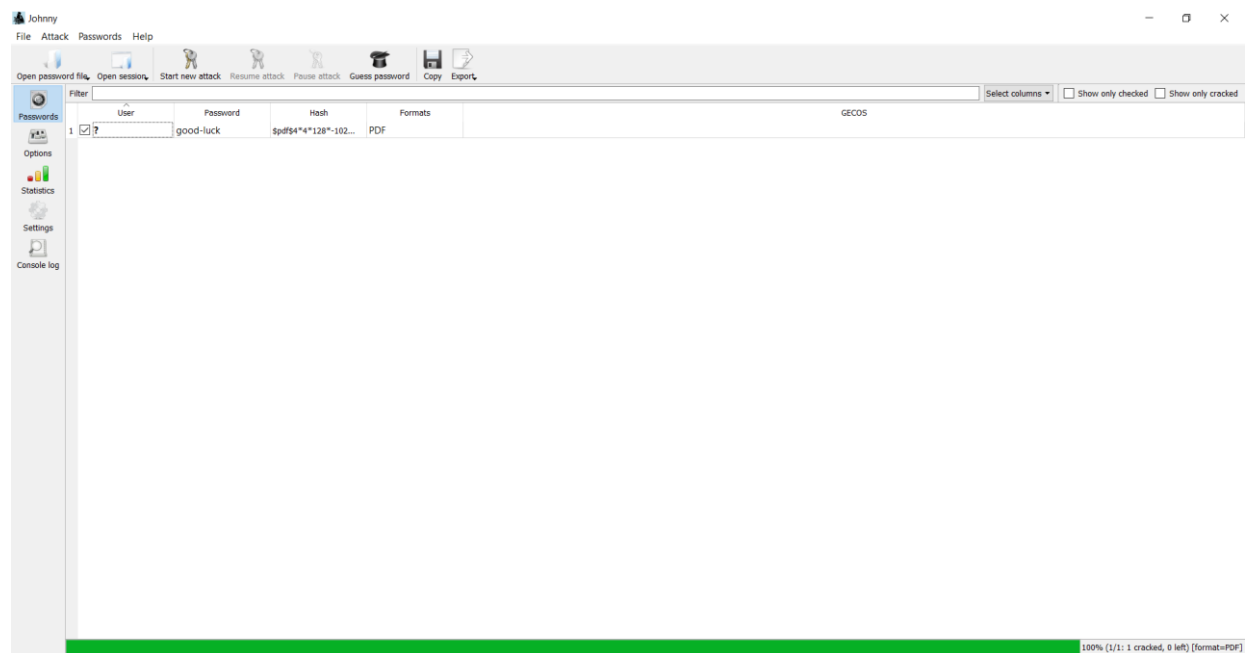
Tools: John the Ripper (jumbo), Johnny GUI (Windows)

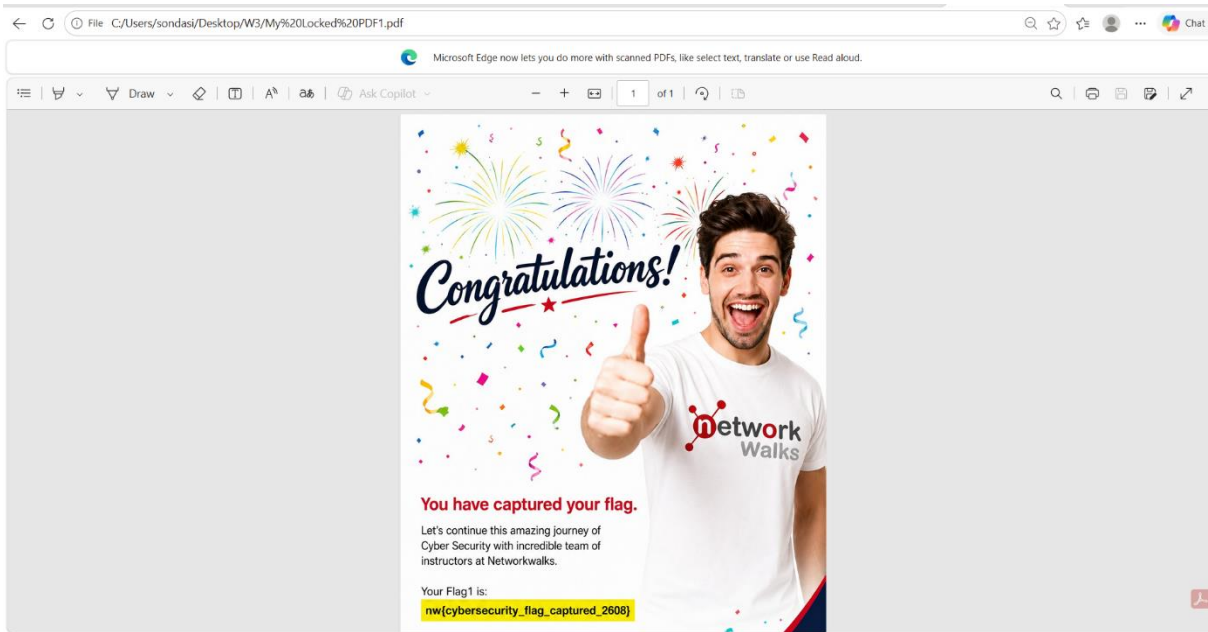
Steps:

1. Installed John the Ripper and Johnny GUI on Windows, pointed Johnny to `john.exe`.
2. Extracted a crackable hash (`\$pdf\$...`) from the locked PDF using an online PDF hash extractor (pdf2john format).
3. Saved the hash into a text file and opened it in Johnny via **Open password file**.
4. Ran **Start new attack** — Johnny/John cracked the hash in seconds.
5. Opened the PDF with the recovered password and captured the flag.

Result: Password cracked, PDF unlocked.

My PDF 1





My PDF3

Johnny

File Attack Passwords Help

Open password file, Open session, Start new attack, Resume attack, Pause attack, Guess password, Copy, Export

Filter

	User	Password	Hash	Formats	
1	?	1qaz2wsx	spdf54*4*128*-102...	PDF	GEOS

100% (1/1: 1 cracked, 0 left) [format=PDF]

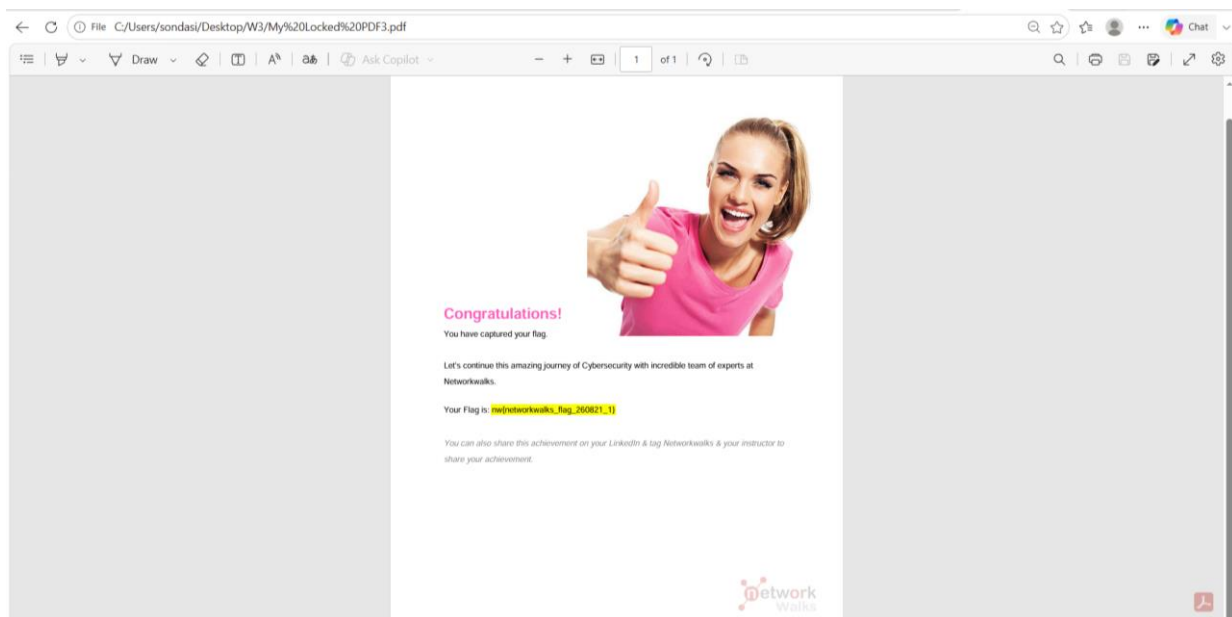
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28°C

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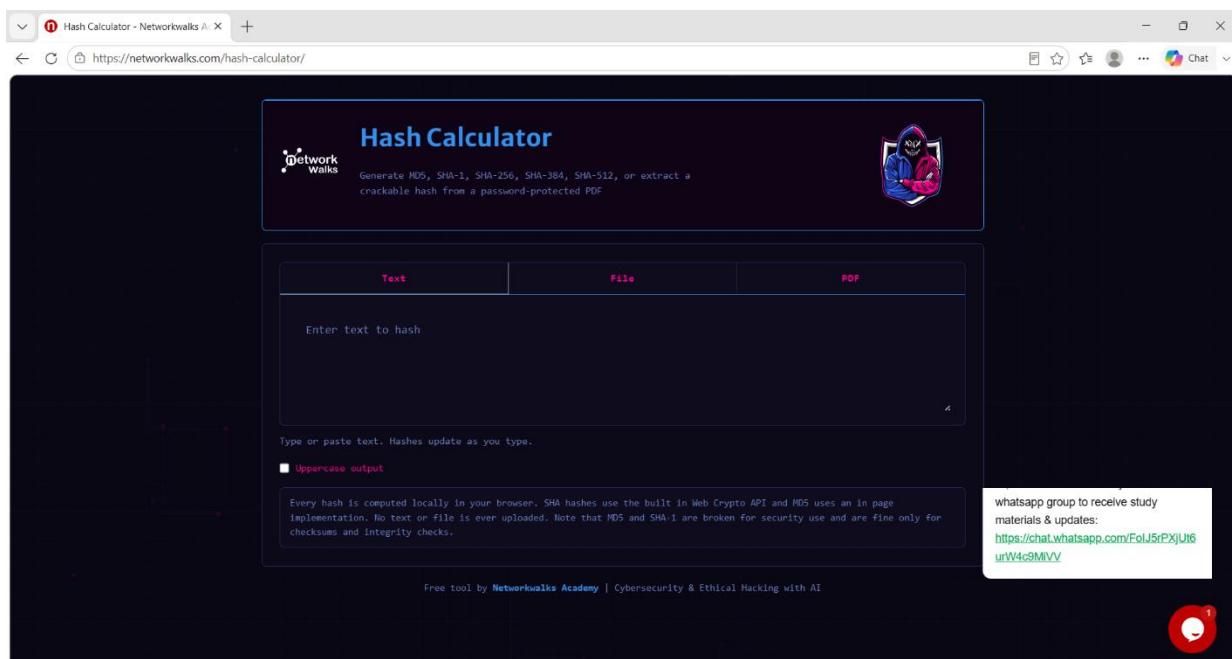


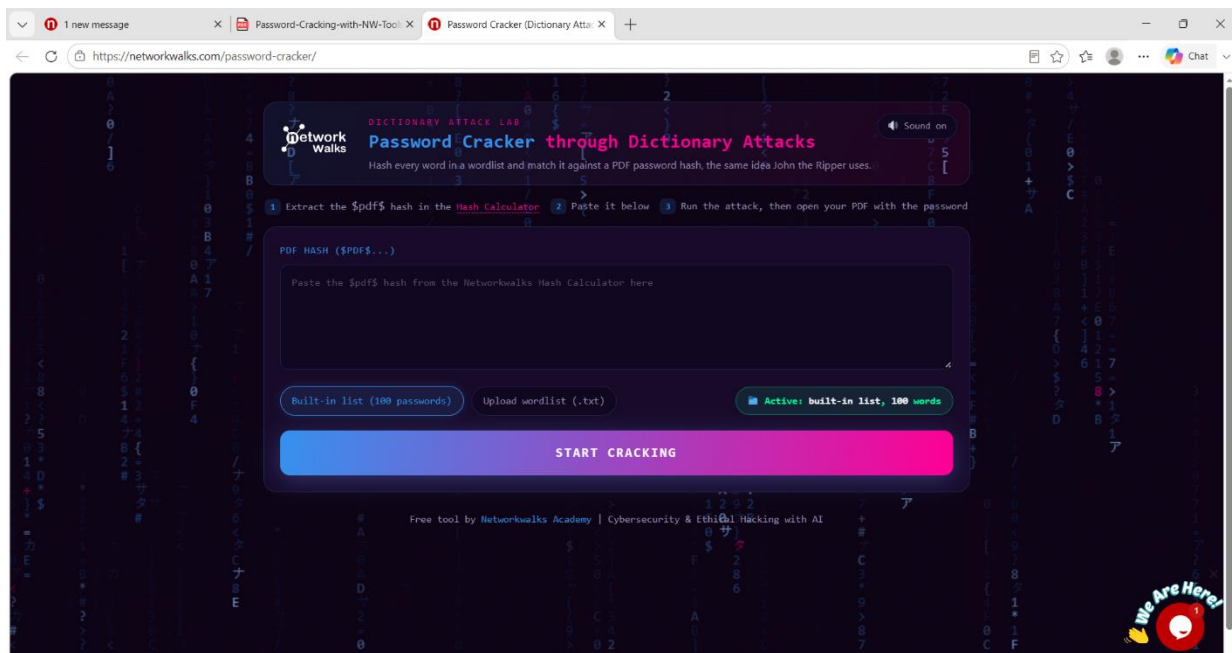
W3-PM2: Password Cracking with NetworkWalks Tools

****Tools:**** NetworkWalks Hash Calculator, NetworkWalks Password Cracker (dictionary attack, browser-based)

****Steps:****

1. Uploaded the locked PDF to the ****Hash Calculator**** (`networkwalks.com/hash-calculator`) to extract its ``$pdf$`` hash locally in-browser.
2. Copied the full hash and pasted it into the ****Password Cracker**** (`networkwalks.com/password-cracker`).
3. Ran the built-in dictionary attack (100-word list) against the hash.
4. Tool matched the password after trying multiple common candidates.
5. Opened the PDF with the cracked password to confirm and capture the flag.

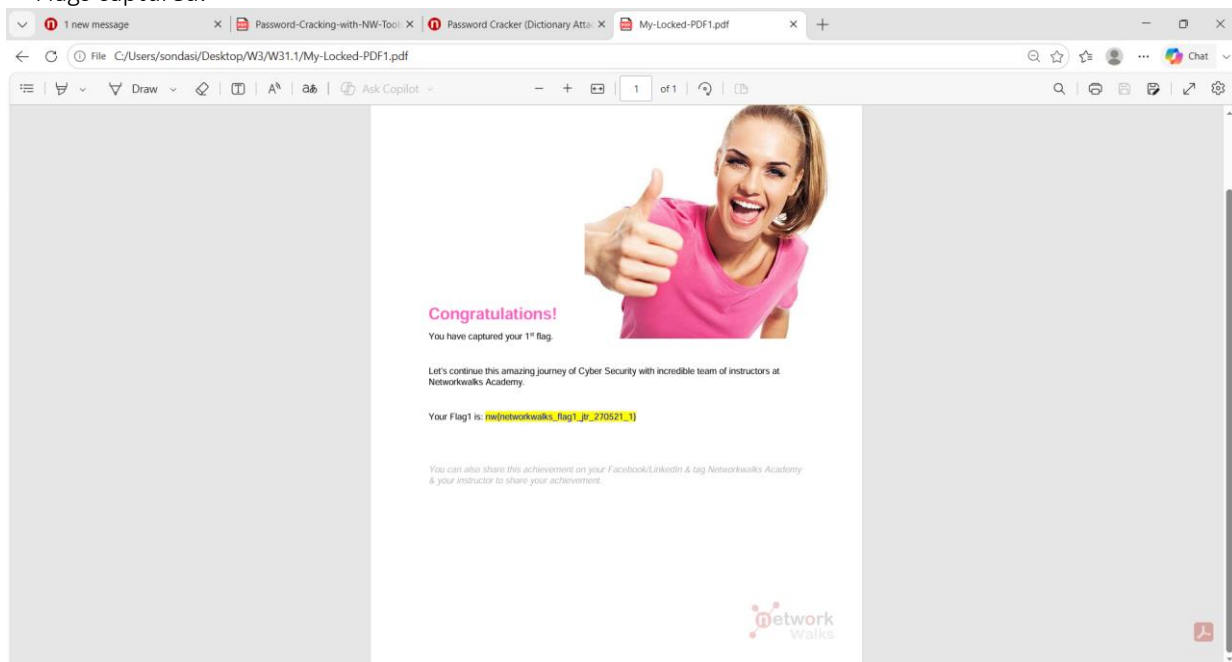


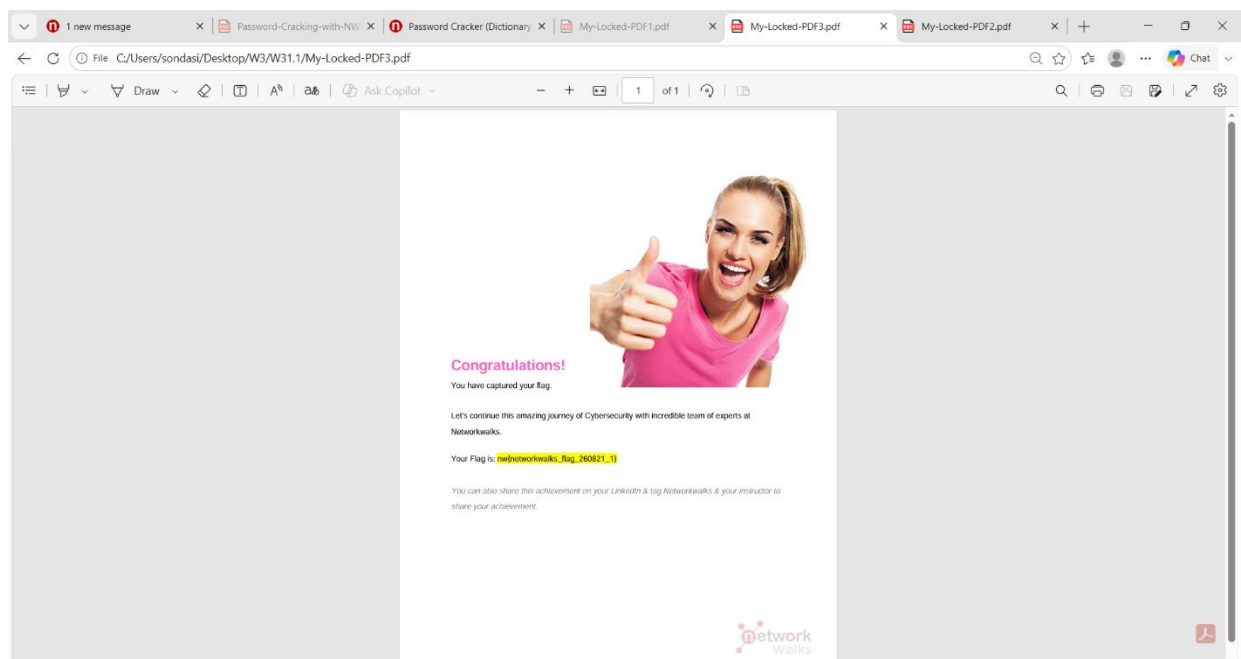
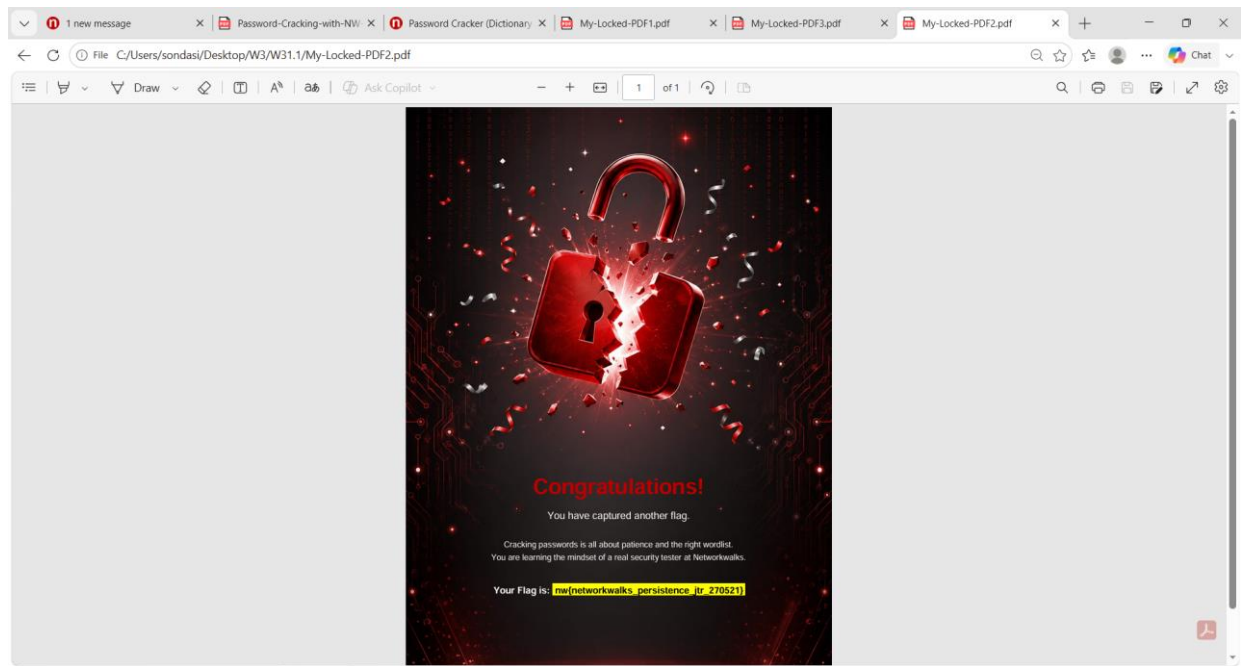


****Results across sample files:****

- `My-Locked-PDF1.pdf`
- `My-Locked-PDF3.pdf`

****Flags captured:****





Key Takeaway

Both a dedicated offline cracking tool (John the Ripper) and a lightweight browser-based dictionary attack were able to recover weak, dictionary-based PDF passwords within seconds — demonstrating why simple/common passwords offer little real protection and why strong, unique passwords matter for encrypted documents.

Disclaimer

This work was performed in a controlled training lab environment provided by NetworkWalks Academy for educational purposes only, using PDF files intentionally created for this exercise.

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